

Kanishka Singh

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🐙 GitHub

📁 Portfolio

Education

New York University - Courant Institute of Mathematical Sciences

Sept 2025 – May 2027

M.S. in Computer Science

New York, USA

Graduating May 2027; Available to start Summer/Fall 2027

Relevant Courses: Machine Learning, Deep Learning, GPUs: Architecture & Programming, Fundamentals of Algorithms

Kalinga Institute of Industrial Technology

Sept 2020 – Apr 2024

B.Tech. in Computer Science & Engineering (GPA: 9.04/10.0)

Bhubaneswar, India

Industry Experience

Ernst & Young, Senior Analyst

Jul 2024 – Apr 2025

- Developed **LLM-powered enterprise search systems** using **OpenAI APIs, LangChain, RAG pipelines, vector retrieval, and GCP-backed data services** for context-aware information access.
- Built and maintained an **LLM-powered conversational search system** with **memory-aware retrieval** and few-shot prompting for multi-turn queries, while debugging, testing, and extending an existing team codebase.
- Built a **machine learning prediction engine** over **2+ years of inventory data**, engineering data-processing and feature pipelines for ensemble models that achieved **85% accuracy** on FMCG stock-out prediction.

Ernst & Young, Data Science & Analytics Intern

Jun 2023 – Jul 2024

- Owned **end-to-end backend design and development** for an **EV analytics platform**, building reusable **Python ETL pipelines**, relational data models, and PostgreSQL views to process large-scale automotive data and support **15 dashboards**.
- Reduced a client dashboard's **refresh time from 60 minutes to 6 minutes** by optimizing complex **PostgreSQL queries** and restructuring backend data retrieval.
- Engineered reusable **Python and SQL data pipelines** over automotive and FMCG datasets with **1M+ row tables**, automating ingestion, batch processing, validation, deduplication, and incremental processing.
- Built an **embedding-based KPI discovery system** indexing KPIs across hundreds of dashboards and using **vector search** to return the **Top-3 relevant dashboard matches** for internal users.

Projects

Multimodal Video Search & Ranking Engine

Python, PyTorch, OpenCLIP, Whisper, BLIP, FAISS, BM25

- Developed a **multimodal video search engine** over transcripts, frames, OCR, and captions using **OpenCLIP, Whisper, and BLIP**.
- Built a **hybrid retrieval and reranking pipeline** with **FAISS, BM25, and Cross-Encoders**; served via **FastAPI, Redis, PostgreSQL, and Docker** and evaluated **Recall@K, MRR, NDCG, and latency**.

Relay: Distributed Job Execution Platform

Go, gRPC, PostgreSQL, Redis, Docker, Kubernetes

- Engineered a **distributed job platform** processing **3,000+ jobs/min across 8 workers** with **<150 ms p95 scheduling latency** using **Go, gRPC, PostgreSQL, and Redis**.
- Added **fault tolerance** with heartbeats, job leases, idempotent retries, and exponential backoff, enabling **<5-second worker recovery**; deployed with **Docker and Kubernetes**.

Personalized Search Ranking Engine

Python, LightGBM, FastAPI, Redis, PostgreSQL, Docker

- Trained a **LambdaMART/LightGBM ranker** using **watch history, engagement, creator affinity, and semantic relevance** to personalize video search.
- Built a **ranking and serving pipeline** with feature engineering, negative sampling, and online features via **Redis/PostgreSQL**; served through **FastAPI** and evaluated **NDCG@K, MRR, MAP, and Recall@K**.

IndexForge: High-Performance Search Engine

C++20, CMake, Multithreading, BM25, GoogleTest

- Implemented a **multithreaded C++20 search engine** over **500K+ documents** with an **inverted index, BM25 ranking, and Top-K retrieval**, achieving **<50 ms p95 latency**.
- Parallelized indexing and queries with **thread pools, synchronization primitives, and cache-aware optimizations**, improving **throughput by 2.5x+** and reducing **p95 latency by 20%+**.

Skills

Languages: Python, C++, Go, Java, SQL, JavaScript, TypeScript

Machine Learning: PyTorch, Scikit-learn, LightGBM, Learning-to-Rank, Recommendation Systems, Feature Engineering

Search & Retrieval: FAISS, BM25, Embeddings, Vector Search, Semantic Search, Cross-Encoders

Backend & Data: FastAPI, gRPC, PostgreSQL, Redis, REST APIs, ETL

AI Systems: LangChain, RAG, OpenAI APIs

Infrastructure: Docker, Kubernetes, AWS, GCP, Git, GitHub Actions, CI/CD